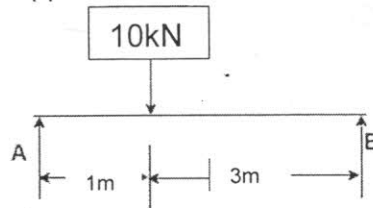




Summer Semester July 2024

Duration: 3 Hours

10. The rate of change of shear force is equal to _____
 A) direction of load B) change in BMD
 C) intensity of loading D) maximum bending
11. What is the maximum shear force, when a cantilever beam is loaded with UDL throughout?
 A) $w \times l$ B) w
 C) w / l D) $w + l$
12. Find the reaction (in kN) at the simple support A?



- A) 6.5 B) 9
 C) 10 D) 7.5
13. Curvature of the beam is _____ to bending moment.
 A) equal B) directly proportional
 C) inversely proportional D) coincident
14. What is the section modulus of rectangle beam of width 240 mm and depth 400 mm? (in terms of 10^6 mm^3)
 A) 5.4 B) 6.2
 C) 5.5 D) 6.4
15. A beam of uniform strength has
 A) same cross-section throughout the beam B) same bending stress at every section
 C) same bending moment at every section D) same shear stress at every section
16. The transverse shear stress acting in a beam of rectangular cross-section, subjected to a transverse shear load, is:
 A) Variable with maximum at the bottom of the beam B) Variable with maximum at the top of the beam
 C) Uniform D) Variable with maximum on the neutral axis
17. Slope is maximum at _____ in simply supported beams.
 A) mid span B) throughout
 C) support D) at the point of loading
18. Deflection of a simply supported beam when subjected to UDL is given as _____
 A) $(Wl / 16 EI)$ B) $(Wl^2 / 16 EI)$
 C) $(Wl^3 / 48 EI)$ D) $(5Wl^4 / 384 EI)$
19. The power transmitted (kW) in the shaft at 150 rpm taking torque as 9000 Nm is approximately
 A) 140 B) 160
 C) 180 D) 250
20. What is the polar modulus for solid shaft in terms of diameter D?
 A) $(\pi/16) D^2$ B) $(\pi/12) D^3$
 C) $(\pi/16) D^3$ D) $(\pi/16) D$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Show that the extension produced due to self-weight of a bar of uniform circular section fixed at one end and suspended vertically is equal to half the extension produced by a load equal to self-weight applied at the free end.	8	L*2	1	2
	b) A bar of 30mm diameter is tested in tension. It is observed that when a load of 60 kN is applied, the extension measured over a gauge length of 200mm is 0.1mm and contraction in diameter is 0.004mm. Find the Poisson's ratio and elastic constants young's modulus, Rigidity modulus and Bulk modulus.	8	L3	2	2

2. a) A member ABCD is subjected to point loads P_1 , P_2 , P_3 , P_4 as shown in Fig.2(a). Calculate the force P_2 necessary for equilibrium, if $P_1 = 45$ kN, $P_3 = 450$ kN and $P_4 = 130$ kN. Determine the total deformation of the member assuming the modulus of elasticity to be 2.1×10^5 N/mm².

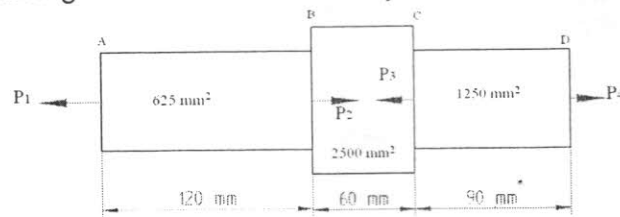


Figure.2(a).

8	L3	1	2
8	L3	2	1

- b) Establish a relation between Modulus of elasticity and rigidity modulus.
3. a) For the laboratory tested specimen the following data were obtained. Diameter of specimen = 30mm, Length of specimen = 200mm, Extension under the load of 150 kN = 0.21mm, Load at yield point = 250 kN, Maximum load = 380 kN, Total extension = 60mm, Neck diameter at failure = 22.5mm. Determine (i) Young's Modulus (ii) Yield stress (iii) Ultimate stress (iv) Percentage Elongation (v) Percentage reduction in area.
- b) A cylindrical shell is 3m long, has internal diameter 1m and 15mm thickness. Calculate the maximum intensity of shear stress induced and also changes in the dimensions of the shell, if it is subjected to an internal fluid pressure of 1.5MPa. Assume modulus of Elasticity as 200GPa and Poisson's ratio as 0.3.

8	L3	1	2
---	----	---	---

8	L3	2	2
---	----	---	---

Unit – II

4. a) Draw shear force and bending moment diagram for the beam as shown below in fig.4(a). Locate point of contra flexure. Show the calculations of shear force and bending moment clearly at each and every point of beam.

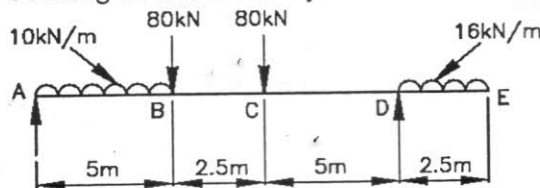


Figure.4(a)

16	L3	3	2
----	----	---	---

5. a) Classify beams and loads with sketches.
- b) Derive the relationship between load, shear force and bending moment.
- c) Prove that the maximum shear stress is $4/3$ times the average shear stress in circular cross section of diameter d .
6. a) Derive a relationship between bending stress, radius of curvature and moment of inertia.
- b) The T-section shown in fig.6(b). below is used as a simply supported beam over a span of 4 meters. It carries a uniformly distributed load of 8 kN/m over its entire span. Calculate the maximum tensile and compressive stresses occurring in the section.

4	L1	3	1
4	L2	3	1

8	L2	4	1
---	----	---	---

8	L2	4	1
---	----	---	---

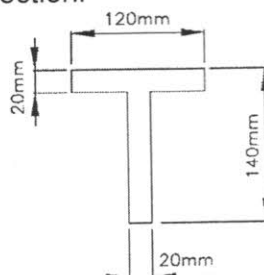


Figure.6(b)

8	L3	4	2
---	----	---	---

Unit – III

7. a) Derive the expression for maximum deflection of a cantilever beam of length 'l' carrying a point load 'W' at the free end.
- b) Find the slope and deflection at the free end of the cantilever shown in fig.7(b). Assume $E=200\text{GPa}$, $I=40\times 10^6\text{ mm}^4$

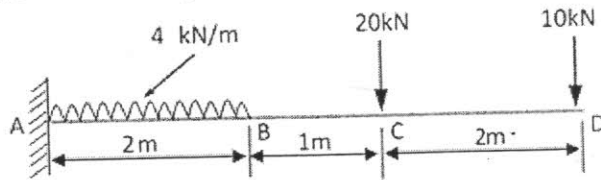


Figure.7(b)

8. a) Determine the diameter of a solid shaft which will transmit 90kW at 160 rpm. The angle of twist should not exceed one degree per meter length and the maximum torsional shear stress is to be limited to 60MPa. Assume $G=80\text{GPa}$.
- b) A solid shaft rotating at 1000 rpm transmits 50 kW. Maximum torque is 20% more than the mean torque. Material of the shaft has the allowable shear stress of 50 MPa and Modulus of rigidity 80 GPa. Angle of the twist in the shaft should not exceed 1° in one-meter length. Determine the diameter of the shaft.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
